EXPERIMENT-3

REG NO-240701164. NAME-HARI KRISHNAN EP

USER INTERFACE DESIGN

1.COMMAND LINE INTERFACE

C CODE:

```
#include <stdio.h>

int main(int argc, char *argv[]) {
    if (argc != 3) {
        printf("Usage: %s <old_filename> <new_filename>\n", argv[0]);
        return 1;
    }

    // Rename the file
    if (rename(argv[1], argv[2]) == 0) {
        printf("File renamed from %s to %s\n", argv[1], argv[2]);
    } else {
        perror("Error");
    }

    return 0;
}
```

OUTPUT:

```
Usage: /tmp/7tRCdfedwm/main.o <old_filename> <new_filename>


=== Code Exited With Errors ===
```

2.GRAPHICAL USER INTERFACE:
C CODE:

```
#include <stdio.h>

int main() {

    char oldName[100];
    char newName[100];
    FILE *fp;

    printf("File Renamer\n");
    printf("============\n\n");

    printf("Enter Old Filename: ");
    scanf("%s", oldName);

    printf("Enter New Filename: ");
    scanf("%s", newName);

    // Check if file exists
    fp = fopen(oldName, "r");

    if (fp == NULL) {
        printf("\nError: File '%s' not found!\n", oldName);
        return 1;
    }

    fclose(fp);

    // Rename file
    if (rename(oldName, newName) == 0) {
        printf("\nSuccess!\n");
        printf("File renamed from %s to %s\n", oldName, newName);
    } else {
        printf("\nError: Unable to rename file!\n");
```

```
}

return 0;
```

```
File Renamer
=============

Enter Old Filename: HK.TXT
Enter New Filename: MAHA.TXT
ERROR!

Error: File 'HK.TXT' not found!


=== Code Exited With Errors ===
```

3.VOICE USER INTERFACE:

PYTHON CODE:

```
import speech_recognition as sr
import os


def rename_file_from_voice(command):
    try:
        words = command.lower().split()

        # Check format: rename <old> to <new>
        if "rename" in words and "to" in words:

            rename_index = words.index("rename")
            to_index = words.index("to")

            old_name = words[rename_index + 1]
            new_name = words[to_index + 1]

            # Check if old file exists
            if not os.path.exists(old_name):
                print(f"❌ Error: File '{old_name}' not found.")
                return

            # Rename file
            os.rename(old_name, new_name)
            print(f"✅ File renamed from '{old_name}' to '{new_name}'")

        else:
            print("❌ Invalid command format.")
            print("Say: Rename oldfile.txt to newfile.txt")

    except Exception as e:
        print(f"❌ Error: {e}")
```

```
def listen_for_command():

    recognizer = sr.Recognizer()
    mic = sr.Microphone()

    print("🎤 Listening for command to rename a file...")

    with mic as source:
        recognizer.adjust_for_ambient_noise(source)
        audio = recognizer.listen(source)

    try:
        command = recognizer.recognize_google(audio)

        print(f"🗣️ Command received: {command}")

        rename_file_from_voice(command)

    except sr.UnknownValueError:
        print("❌ Sorry, I couldn't understand.")

    except sr.RequestError as e:
        print(f"❌ API Error: {e}")


if _name_ == "_main_":
    listen_for_command()
```

OUTPUT:

```
Traceback (most recent call last):
  File "/home/main.py", line 1, in <module>
    import speech_recognition as sr
ModuleNotFoundError: No module named 'speech_recognition'


...Program finished with exit code 1
Press ENTER to exit console.
```

4.USER SATISFACTION COMPARISION:
C CODE:

```
#include <stdio.h>

int main() {

    int cli, gui, vui;
    float average;

    printf("Rate your satisfaction with the following interfaces (1-5):\n");

    // Get user input
    printf("CLI (1-5): ");
    if (scanf("%d", &cli) != 1) {
        printf("Invalid input! Please enter numbers only.\n");
        return 1;
    }

    printf("GUI (1-5): ");
    if (scanf("%d", &gui) != 1) {
        printf("Invalid input! Please enter numbers only.\n");
        return 1;
    }

    printf("VUI (1-5): ");
    if (scanf("%d", &vui) != 1) {
        printf("Invalid input! Please enter numbers only.\n");
        return 1;
    }

    // Validate range (1 to 5)
    if (cli < 1 || cli > 5 || gui < 1 || gui > 5 || vui < 1 || vui > 5) {
        printf("Please enter ratings between 1 and 5 only.\n");
        return 1;
    }
```

```
    // Display ratings
    printf("\nYour satisfaction ratings:\n");
    printf("CLI: %d\n", cli);
    printf("GUI: %d\n", gui);
    printf("VUI: %d\n", vui);

    // Calculate average
    average = (cli + gui + vui) / 3.0;

    printf("\nAverage Satisfaction Score: %.2f\n", average);

    return 0;
}
```

OUTPUT:

```
Rate your satisfaction with the following interfaces (1-5):
CLI (1-5): 3
GUI (1-5): 2
VUI (1-5): 5

Your satisfaction ratings:
CLI: 3
GUI: 2
VUI: 5

Average Satisfaction Score: 3.33
```